

Integrated Municipality Management Information System (IMMIS)

Project Overview and Problem Description

The Integrated Municipality Management Information System (IMMIS) is a web-based platform designed to automate and centralize essential municipal services. Municipalities often rely on manual paperwork for handling certificates, tax payments, complaints, waste management, tenders, and recruitment, resulting in inefficiency, data loss, and long processing times. The IMMIS system solves these issues by providing a unified digital interface for citizens, contractors, and municipal staff. It enhances transparency, reduces human error, and supports modern e-governance practices. **Objectives:**

- Automate administrative workflows (certificate issuance, tax collection, tender posting, etc.)
- Offer an integrated interface for all stakeholders (citizens, contractors, staff, and administrators)
- Ensure secure data management and role-based access
- Improve efficiency in waste collection and complaint resolution
- Promote transparency in municipal job postings and tenders

Scope:

The system serves municipal authorities, citizens (through NID verification), contractors, job seekers, and administrators. It covers modules for certificate issuance, complaints, service requests, tenders, taxes, jobs, waste tracking, and reporting dashboards.

Requirement Elicitation Process

The project requirements were collected through:

- Interviews and discussions with municipal staff to identify operational pain points.
- Citizen feedback about common service delays and difficulties in accessing government offices.
- Observation of manual processes like certificate issuance and complaint tracking.
- Reference study of existing digital governance models used in other cities.

Stakeholders Involved:

Municipal Administrator, Municipal Staff and Clerks, Citizens and Job Seekers, Contractors/Bidders, Waste Management Teams, Tax Officers, IT Team/System Administrator.

Requirement Analysis

Functional Requirements:

- Citizens can apply for certificates, pay taxes, and track complaints.
- Contractors can bid for tenders and view bid results.
- Admins can post jobs, approve applications, and view reports.
- Tax officers can generate bills and verify payments.
- System sends automated email/SMS notifications.

Non-Functional Requirements:

- Role-based authentication for secure access.
- Responsive and mobile-friendly interface.
- Backup and data recovery mechanism.
- Real-time notifications and dashboards.
- Audit trails for accountability.

Use Cases:

Each module (certificate, tax, complaint, tender, job, service request, and waste) has defined actors and workflows. Example: In Certificate Management, a citizen applies → staff verifies → admin

approves → certificate issued and notified via SMS/email.

System Modeling and Diagrams

Use Case Diagram: Shows interactions between key actors (citizens, contractors, staff, admin) and system modules such as tax, job, complaint, and service management. External systems like NID Verification and Payment Gateway ensure identity validation and secure transactions.

Activity and Swimlane Diagrams: Depict workflow steps for each service, assigning responsibilities across user roles. Examples include how certificates are issued, how complaints are processed, and how tenders are reviewed and awarded.

Class Diagram: Displays core classes — Citizen, Admin, Contractor, Municipal Staff, HR, and System — along with their relationships and methods, defining the logical structure and operations of IMMIS.

Data Flow Diagram (DFD): Visualizes data movement within IMMIS. At Level 0, external entities (NID System, Payment Gateway, Email/SMS Service) communicate with IMMIS. At Levels 1–2, each module (Account, Tax, Job, Service, Tender, etc.) is broken into sub-processes showing how data flows between users, databases, and services.

Prototype Design and User Validation

The prototype was developed based on the system diagrams using an iterative design approach.

User Interface: Simple dashboard-based interface for staff, citizens, and administrators.

Testing: Conducted internal testing with simulated data and received feedback from sample users.

Validation: Based on feedback, navigation and form layouts were improved for better usability.

Result: The prototype successfully demonstrated online service requests, tax payment, and notification processes.

User Manual

For Citizens:

1. Login using NID credentials.
2. Apply for certificates, file complaints, or request municipal services.
3. Track application status via dashboard or SMS.
4. Make tax payments securely through the integrated gateway.

For Staff and Admins:

1. Log in with authorized credentials.
2. Approve applications, manage tenders, and monitor system activity.
3. Generate reports from the admin panel.
4. Maintain records, schedule waste management tasks, and handle citizen queries.

Conclusion and Future Work

The IMMIS system successfully integrates multiple municipal functions into a unified online platform. It enhances efficiency, transparency, and service accessibility for both citizens and staff.

Future Enhancements:

- Integration with GIS for real-time service mapping.
- AI-based predictive waste collection scheduling.
- Public feedback analytics dashboard.
- Mobile app version for wider accessibility.

Task Distribution:

Member	ID	Responsibilities
Meherab Hossen Opi	202122104028	System architecture design, diagrams, testing, Prototype development, testing.
Md. Ehasan Mahmud	202122104026	Diagrams, prototype development, user interface, report writing, and testing.
Abdullah Al Mahmud	202122104014	Diagrams, Requirement collection, prototype development, report writing, data modeling, and testing.

Introduction

This project aims to build a web-based **Municipality Management Information System (MMIS)** to streamline essential administrative services and improve accessibility for citizens, contractors, and municipal staff. The system will cover core functions such as **certificate issuance, property tax payment, public service requests, complaint resolution, e-tendering, waste collection tracking, and municipal job postings**.

Citizen data will be verified via the **National ID (NID)**, removing the need for a separate registration process within the system. The inclusion of modules such as **Jobs, Tender Management, Waste Tracking, and Tax Services** supports modern digital governance and boosts transparency.

Objectives

- Automate major municipal administrative workflows
- Offer a unified platform for staff, citizens, and contractors
- Provide secure and transparent job and tender posting features
- Optimize waste management operations across city zones
- Enable online property tax calculation and payment
- Streamline service requests, complaints, and document issuance
- Enhance transparency and reduce manual processing

Scope of the Project

The system will serve municipal officials, staff, citizens (via NID), contractors, and job seekers. It will include:

- Certificate application and issuance
- Complaint and service request tracking
- Online tender posting and contractor bidding
- Waste collection monitoring by geographic zones
- Tax management with real-time billing and payments
- **Job listing module** for publishing municipal job openings
- Dashboards and analytics for decision-makers

Modules Overview

Module	Description
Certificate Services	Issue certificates such as birth/death based on requests
Tax Management	Calculate, view, and pay property taxes
Complaint System	Citizens can file and track complaints
Service Requests	Apply for services (sanitation, road repair, etc.)
Tender Management	Publish tenders, accept bids, evaluate contractors
Waste Management	Track zone-wise collection, assign teams, monitor performance
Job Portal	Post municipal job openings, allow application submission
Reports Dashboard	Visual summaries of system activity and performance

Stakeholders List

Municipal Administrator (Admin)	Overall system oversight, user management, and high-level approvals. ?
Municipal Clerks / Staff	Handle day-to-day administrative tasks, data processing, and citizen support.
Citizens	Primary users who apply for services, pay taxes, file complaints, and request certificates.
Contractors / Bidders	Participate in e-tendering, submit proposals, and view bid results.
Waste Management Staff	Perform and report on waste collection duties across city zones.
Tax Officers / Revenue Dept.	Manage property tax assessments, billing, and revenue tracking.
HR / Recruitment Officers	Post job vacancies, manage applications, and conduct municipal recruitment.
System Administrator / IT Team	Maintain system health, security, backups, and technical support.
City Planners / Zonal Managers	Use analytics and GIS data for planning infrastructure and services in city zones.
Auditors / Government Inspectors	Access system logs and reports for audits, reviews, and compliance monitoring.
Job Seekers / General Public	View and apply for open government positions listed in the job portal.
External Agencies / Integrators	Interact with the system for data exchange, integration, or regulatory purposes.
Project Team / Developers	Design, develop, test, and update system modules during and after deployment.

Quality Function Deployment (QFD) Requirements

Normal Requirements	
Certificate Issuance	Citizens can request and receive birth, death, and other municipal certificates online.
Complaint System	Allows citizens to file complaints and track their resolution status.
Property Tax Payment	Enables online calculation and payment of property taxes
Tender Posting and Bidding	Municipal authorities can post tenders, and contractors can submit bids
Job Listings and Application	Municipality posts government job openings, and job seekers apply online
NID-Based Login	Citizens authenticate using their National ID, removing the need for separate account creation.
Waste Collection Monitoring	Tracks waste collection activities across city zones.

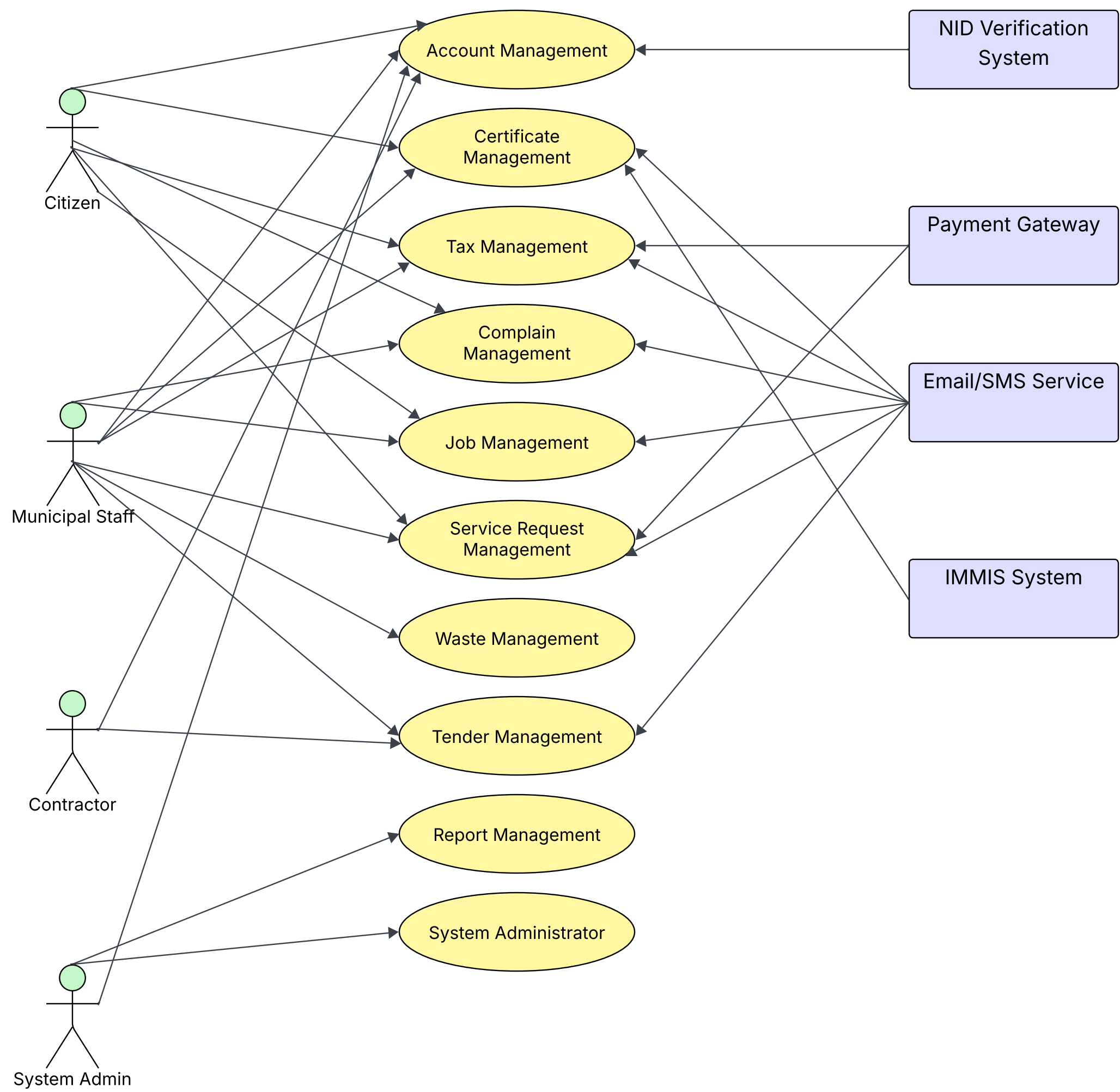
Expected Requirements	
Secure Access & Role-Based Authentication	Ensures data privacy and system integrity for different user roles (admin, citizen, contractor, etc.).
Mobile Compatibility	System functions well on smartphones and tablets.
Audit Trails and Logging	Records every action for security, accountability, and compliance.
Performance Dashboards for Admins	Graphical dashboards to monitor key metrics like complaints, revenue, service status.
Backup and Recovery Mechanism	Periodic data backups and recovery options in case of failure or data loss.
Notifications and Alerts	SMS or email alerts for deadlines, application status, or new service updates.

Exciting Requirements	
GIS-Based Service Mapping	Real-time maps for visualizing infrastructure, complaints, and waste collection zones.
Predictive Waste Scheduling	Uses analytics to optimize waste collection schedules based on past data.
Citizen Feedback Analytics	Collects, analyzes, and displays public satisfaction ratings for various services.
Open API for External Integration	Enables third-party applications or services to interact with IMMIS securely.
Eco Impact Dashboard	Displays metrics like area cleanliness index, recycling rates, and air quality for public awareness.

Use Case Diagram

Level 1: IMMIS Overview

Primary Actors: Citizen, Contractor, Municipal Staff, System Administrator
Secondary Actors: NID Verification System, Payment Gateway, Email Service, SMS Service, IMMIS System

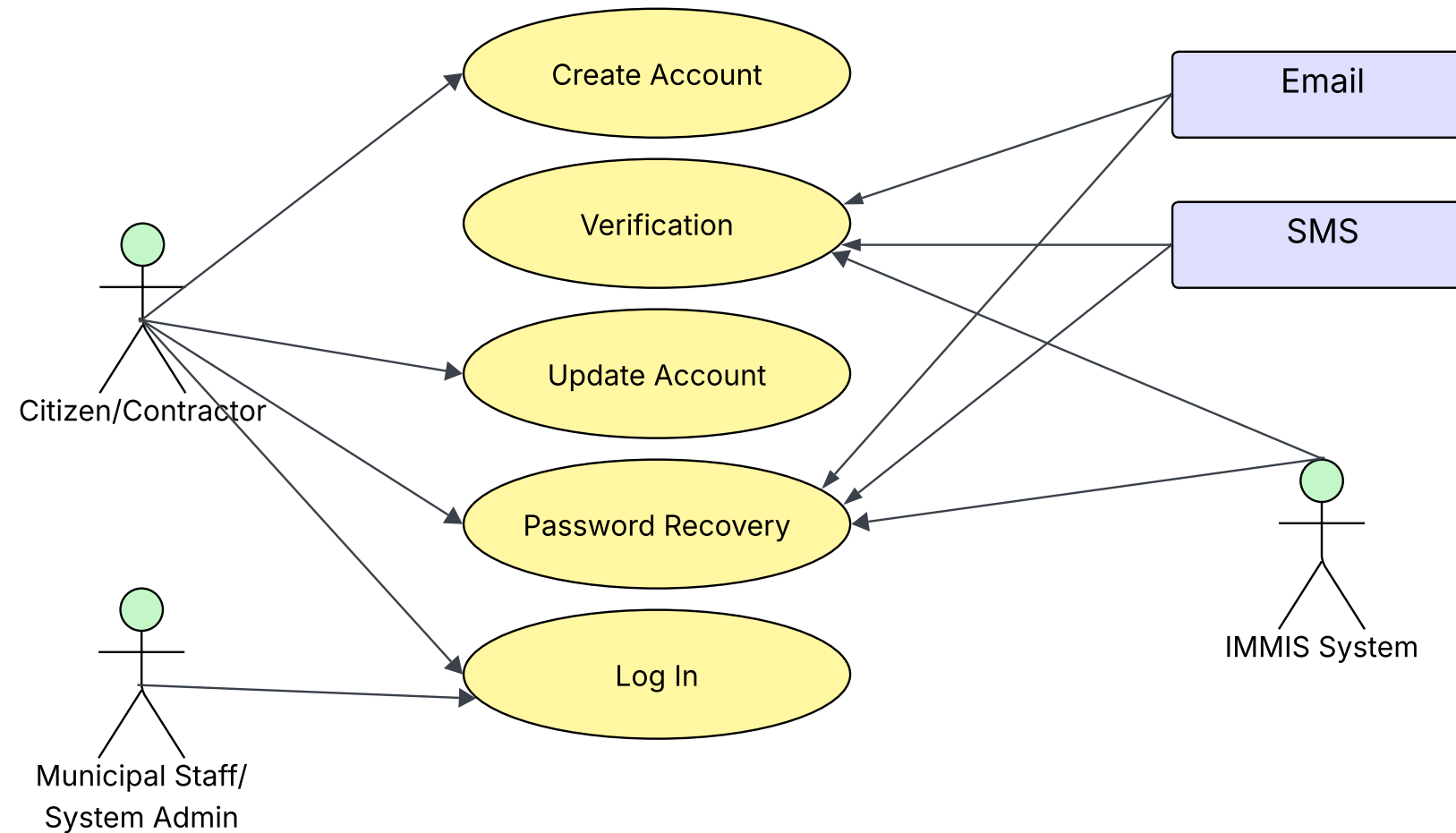


Description: A web-based platform for managing municipal services, enabling citizens, contractors, job seekers, and staff to interact with the system, supported by reporting and administrative tools.

Level 1.1: Account Management

Primary Actors: Citizen, Contractor, Municipal Staff, System Administrator

Secondary Actors: Email Service, SMS Service, IMMIS System



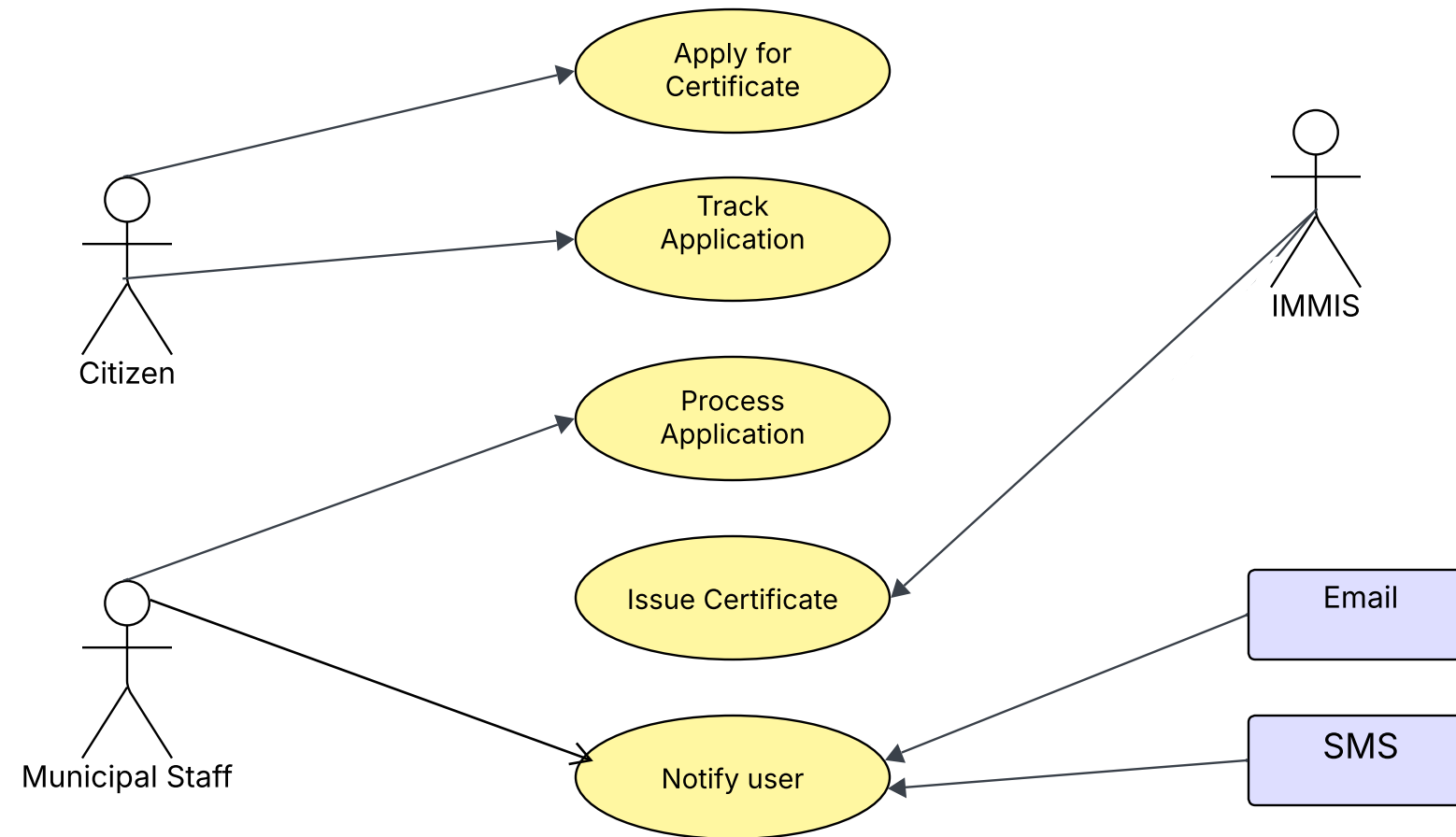
Description: Citizen and Contractor can create a new account and verify or recover their password using Email or SMS. On the other hand, Municipal Staff/ System Admin can Login to the system.

Level 1.2: Certificate Management

Primary Actors: Citizen, Municipal Staff

Secondary Actors: Email Service, SMS Service

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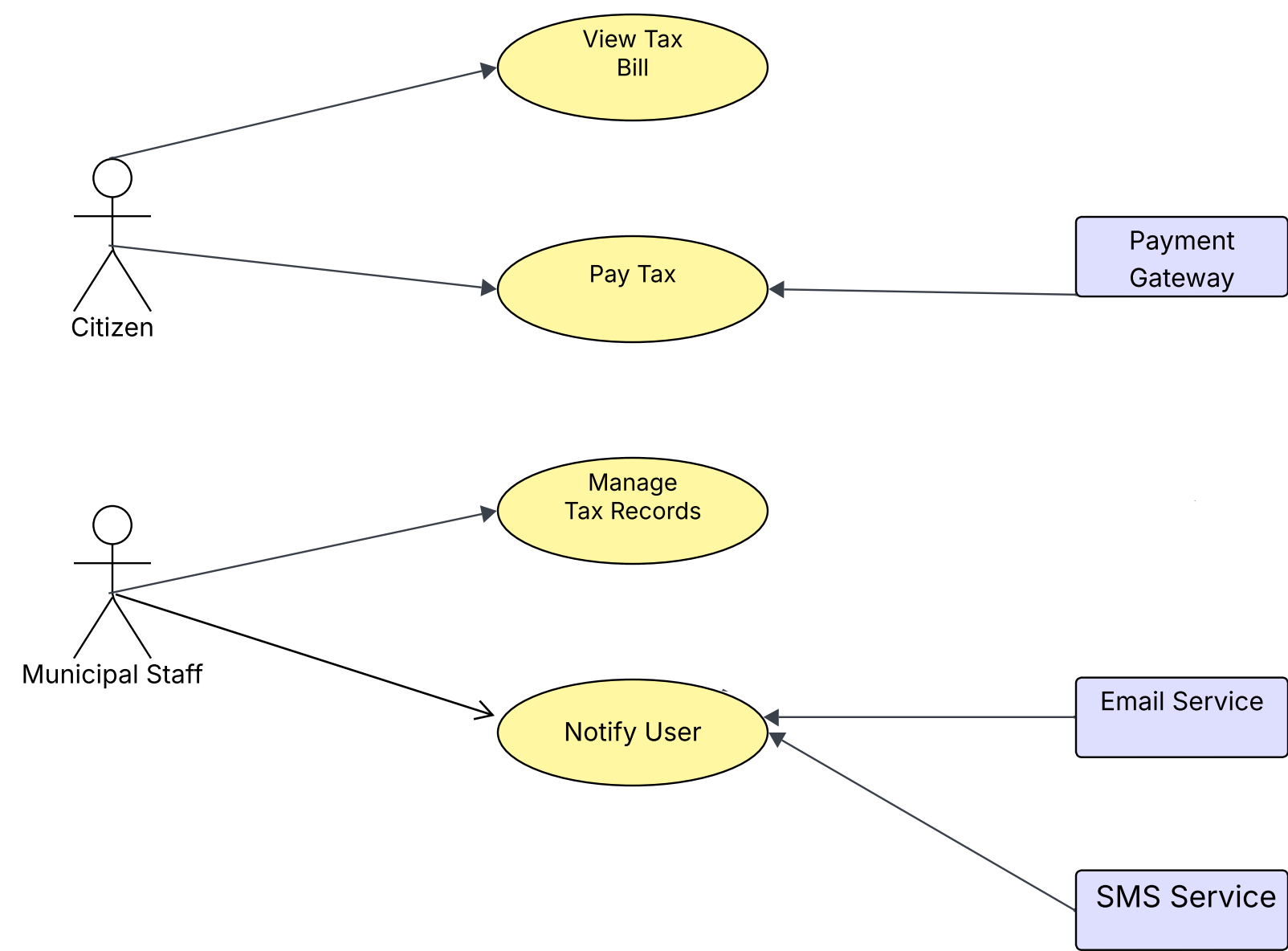


Description: Allows citizens to apply for and track certificates, with staff processing and issuing them, notifying via email / SMS

Level 1.3: Tax Management

Primary Actors: Citizen, Municipal Staff

Secondary Actors: Payment Gateway, Email Service, SMS Service

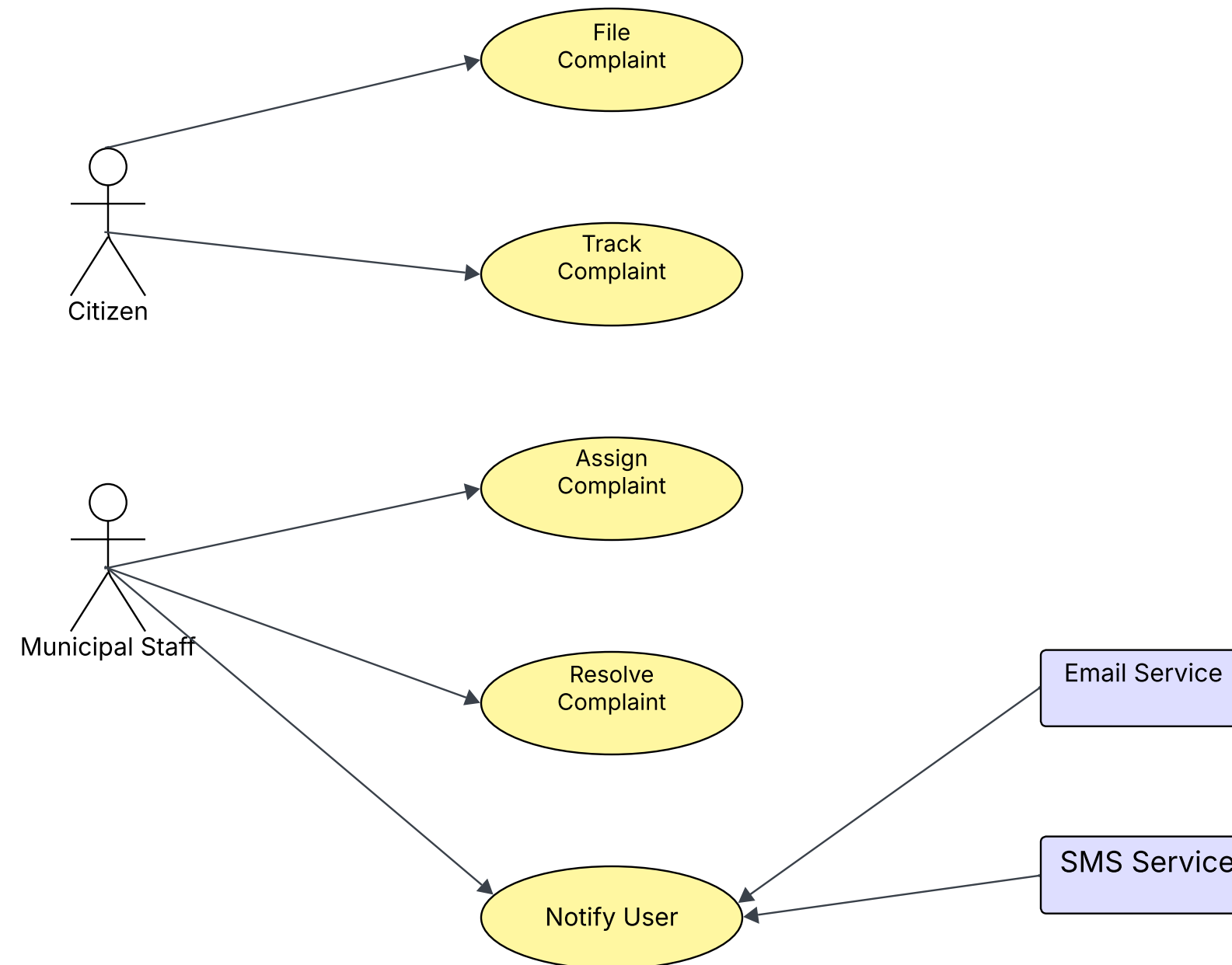


Description: Enables citizens to view and pay taxes online, with staff managing records and payments processed via a gateway, confirmed by email/SMS.

Level 1.4: Complaint Management

Primary Actors: Citizen, Municipal Staff

Secondary Actors: Email Service, SMS Service

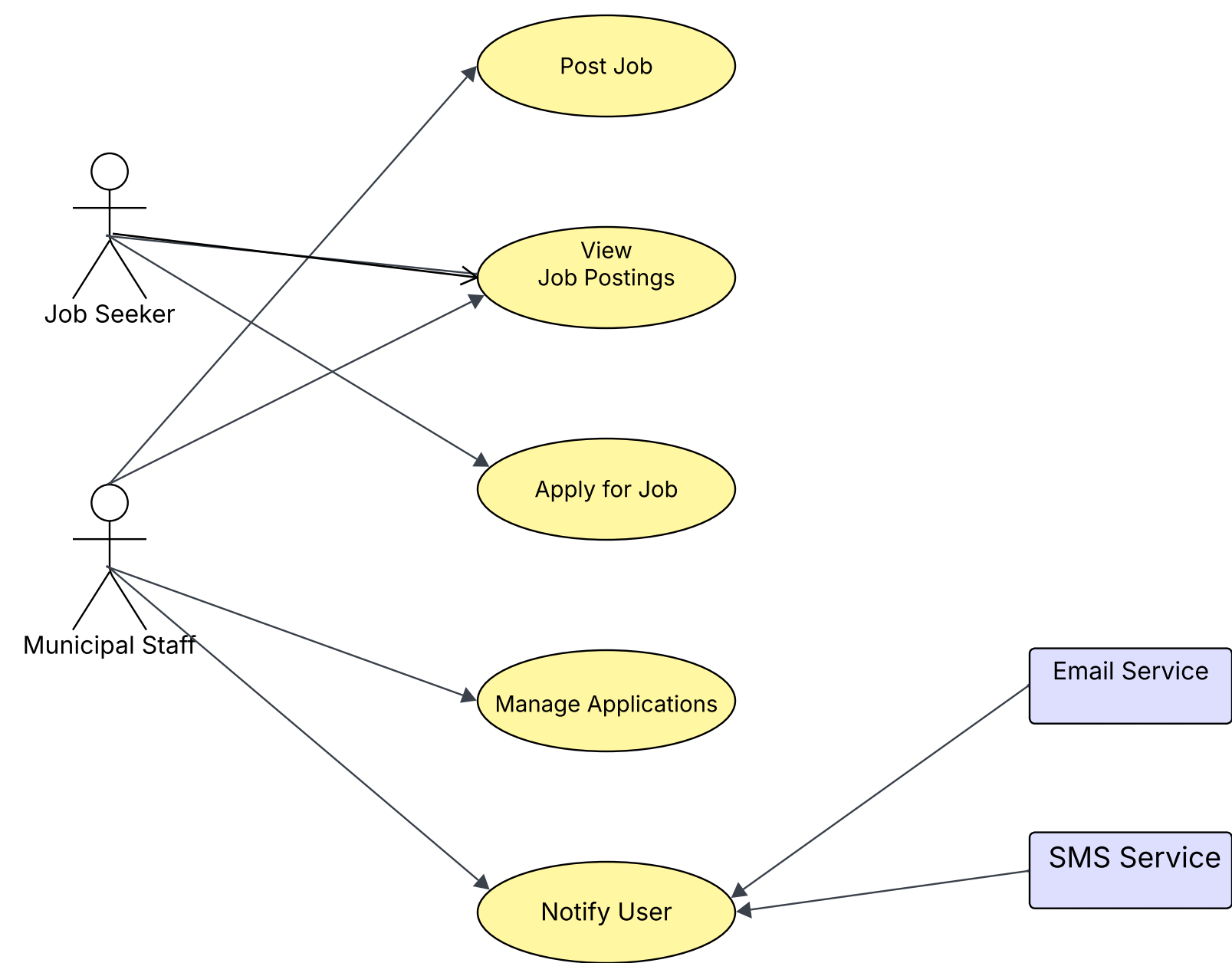


Description: Citizens file and track complaints, staff assign and resolve them, with notifications sent via email/SMS.

Level 1.5: Job Management

Primary Actors: Job Seeker, Municipal Staff

Secondary Actors: Email Service, SMS Service

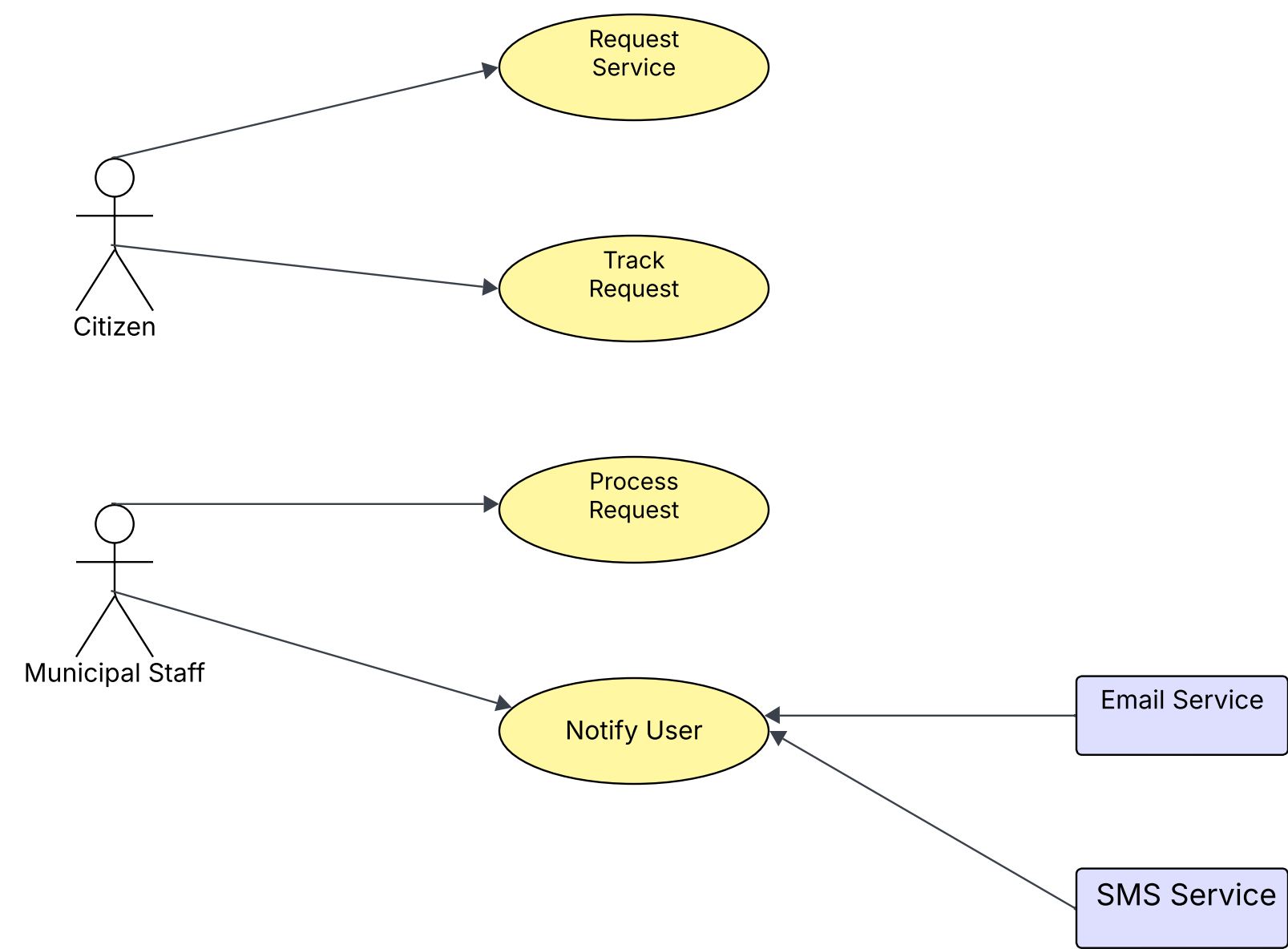


Description: Staff post jobs, seekers apply, and staff manage applications, with email/SMS updates.

Level 1.6: Service Request Management

Primary Actors: Citizen, Municipal Staff

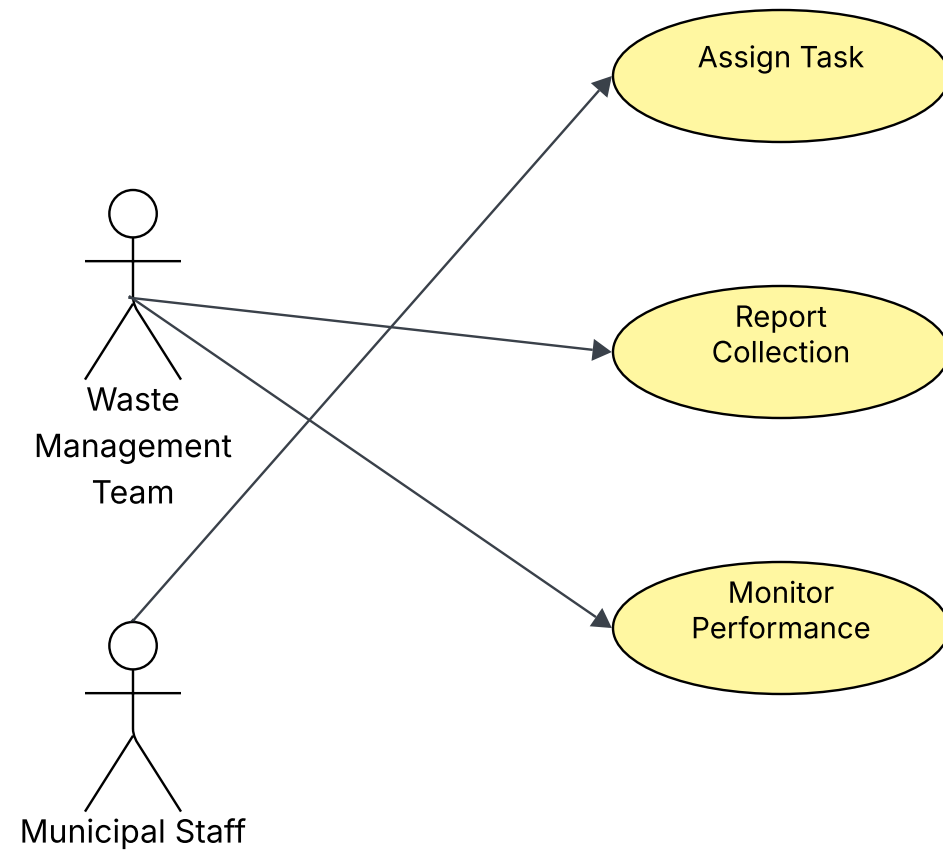
Secondary Actors: Email Service, SMS Service



Description: Citizens request and track services, staff process them, with email/SMS updates.

Level 1.7: Waste Management

Primary Actors: Waste Management Team, Municipal Staff

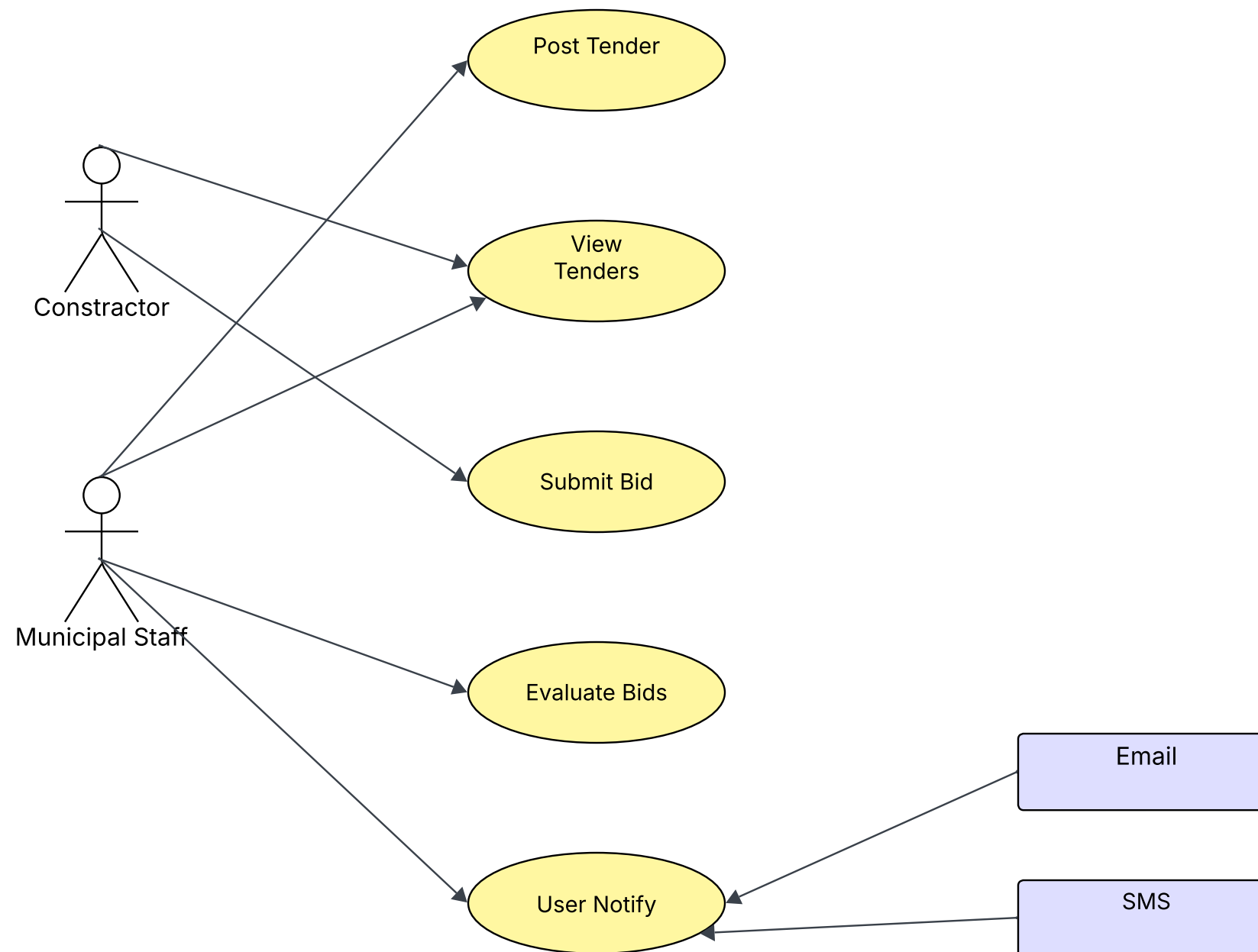


Description: Staff assign tasks, teams report collections, and performance is monitored.

Level 1.8: Tender Management

Primary Actors: Contractor, Municipal Staff

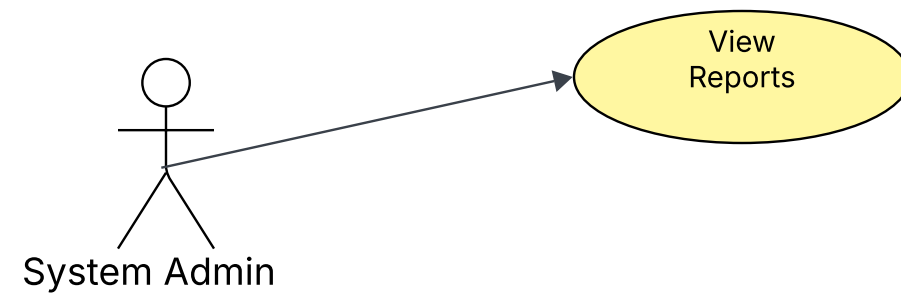
Secondary Actors: Email Service, SMS Service



Description: Staff post tenders, contractors bid, and staff evaluate, with email/SMS notifications.

Level 1.9: Report Management

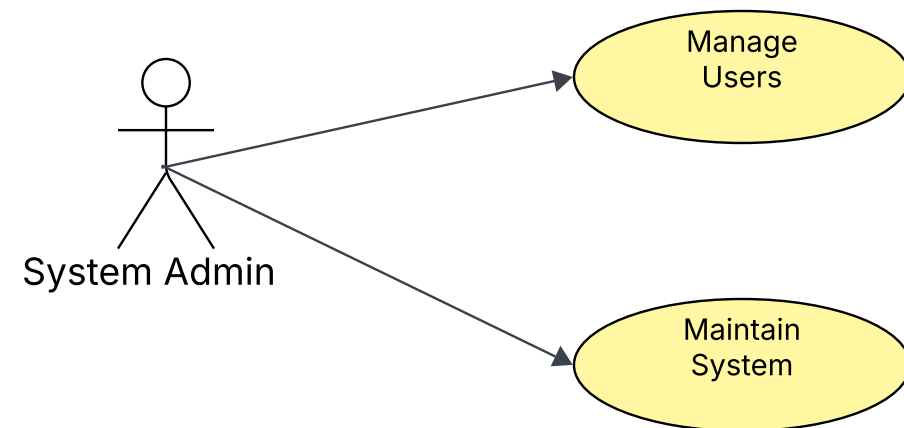
Primary Actors: System Admin



Description: System admin view reports for decision-making.

Level 1.10: System Administration

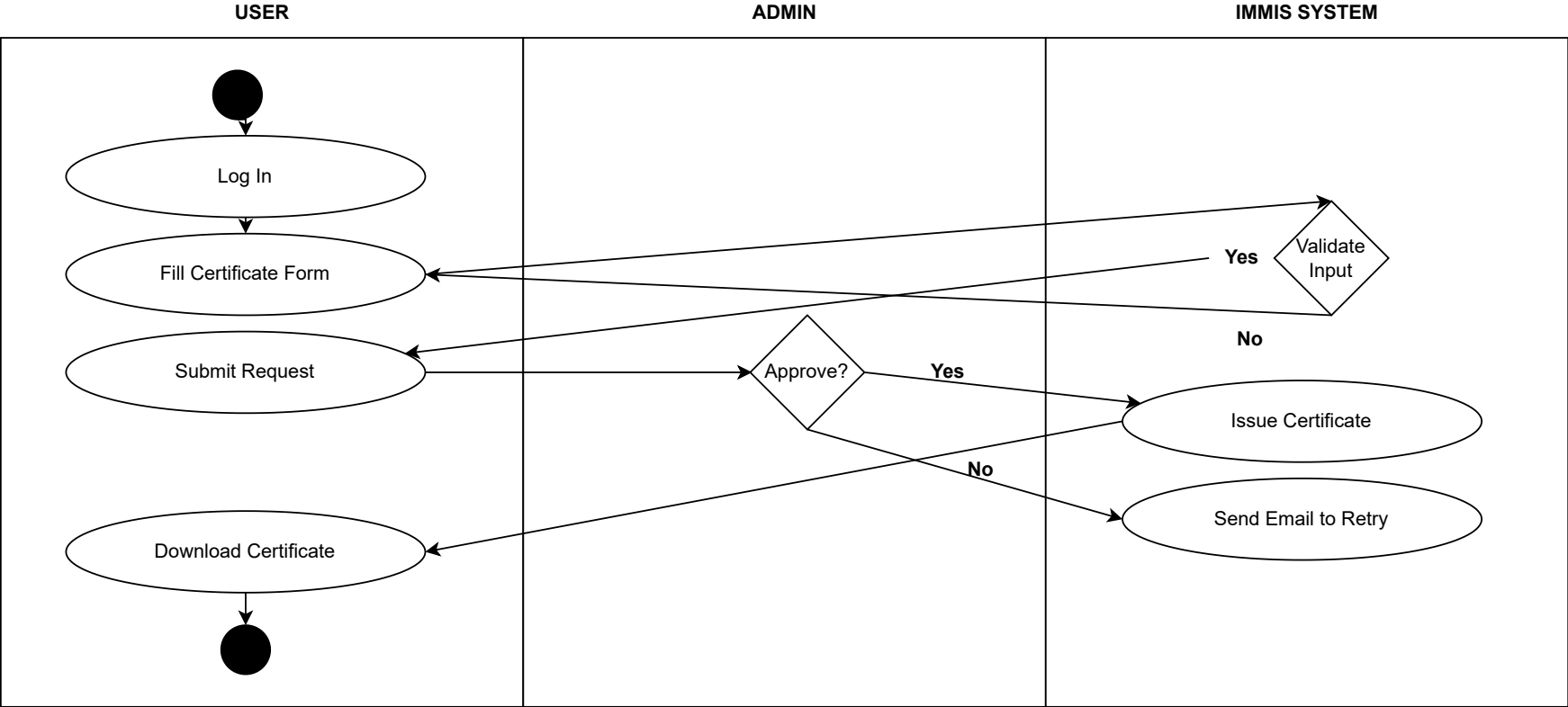
Primary Actors: System Admin



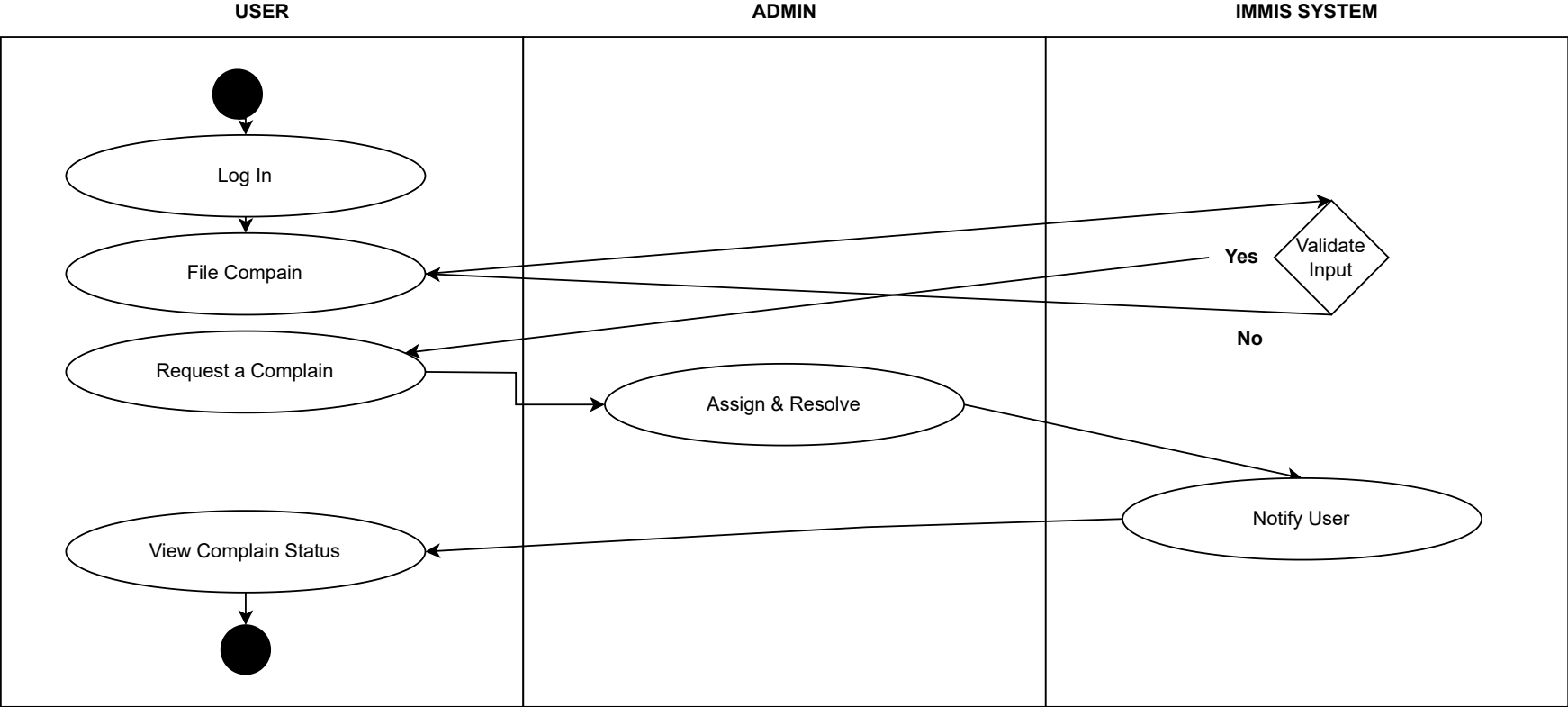
Description: Admins manage users and maintain the system.

Activity and Swimlane Diagram

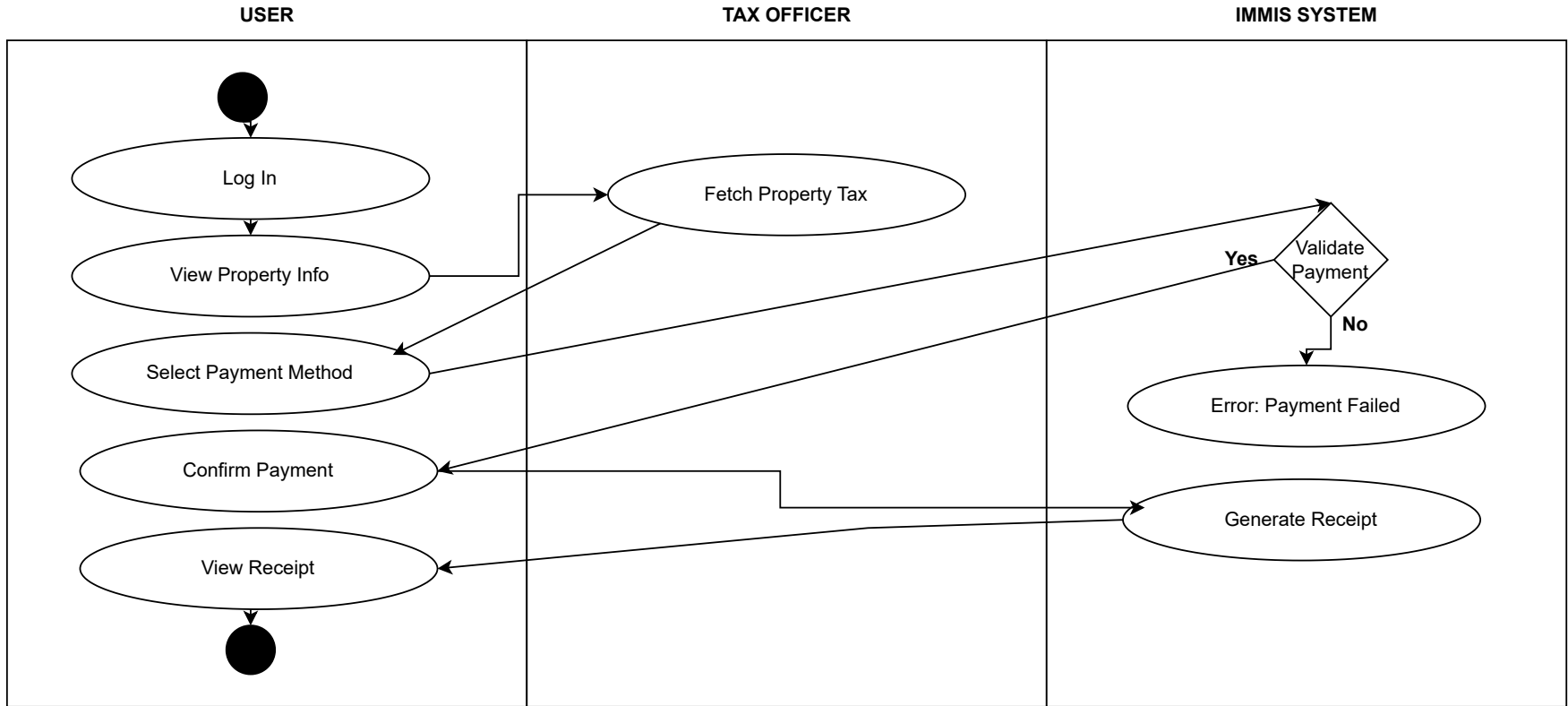
1. NID Service



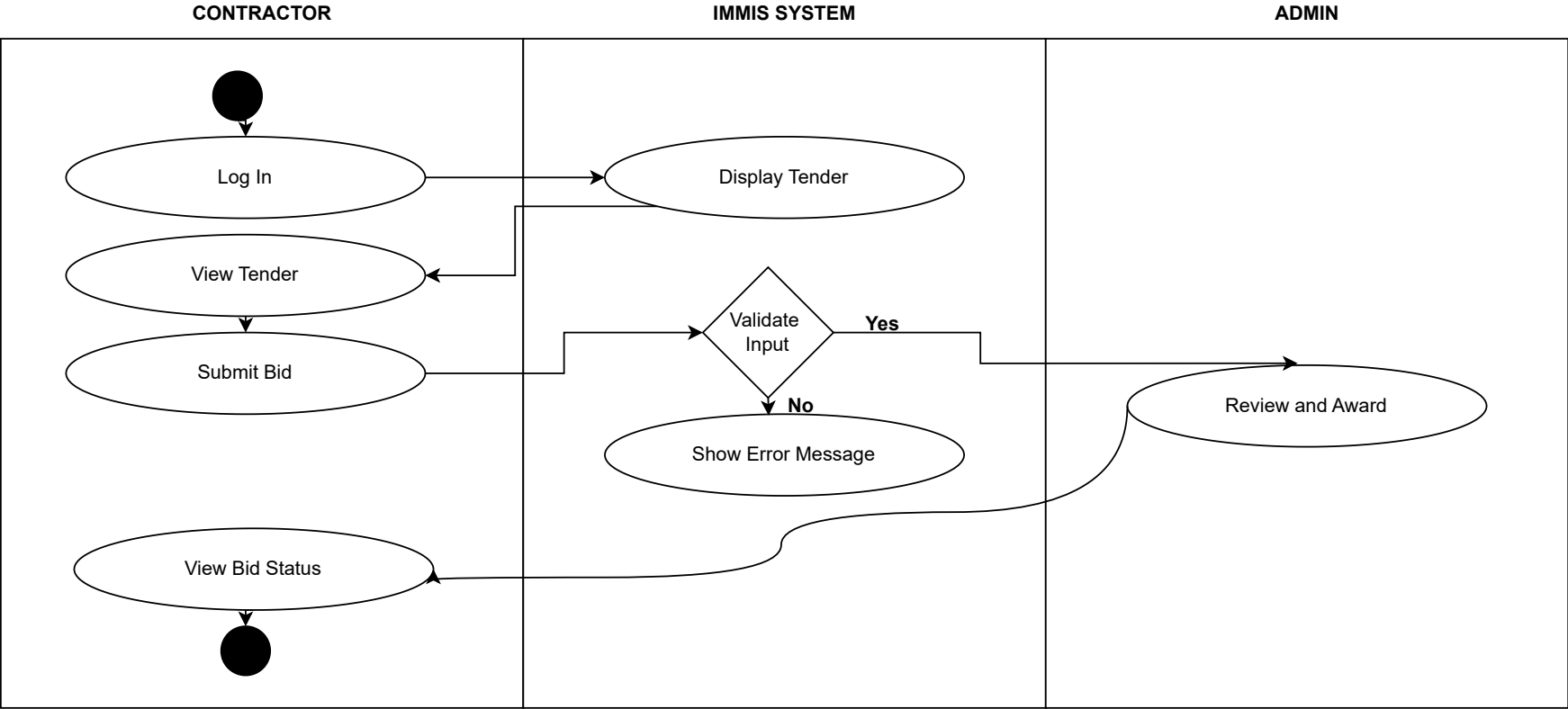
2. Complain Management Service



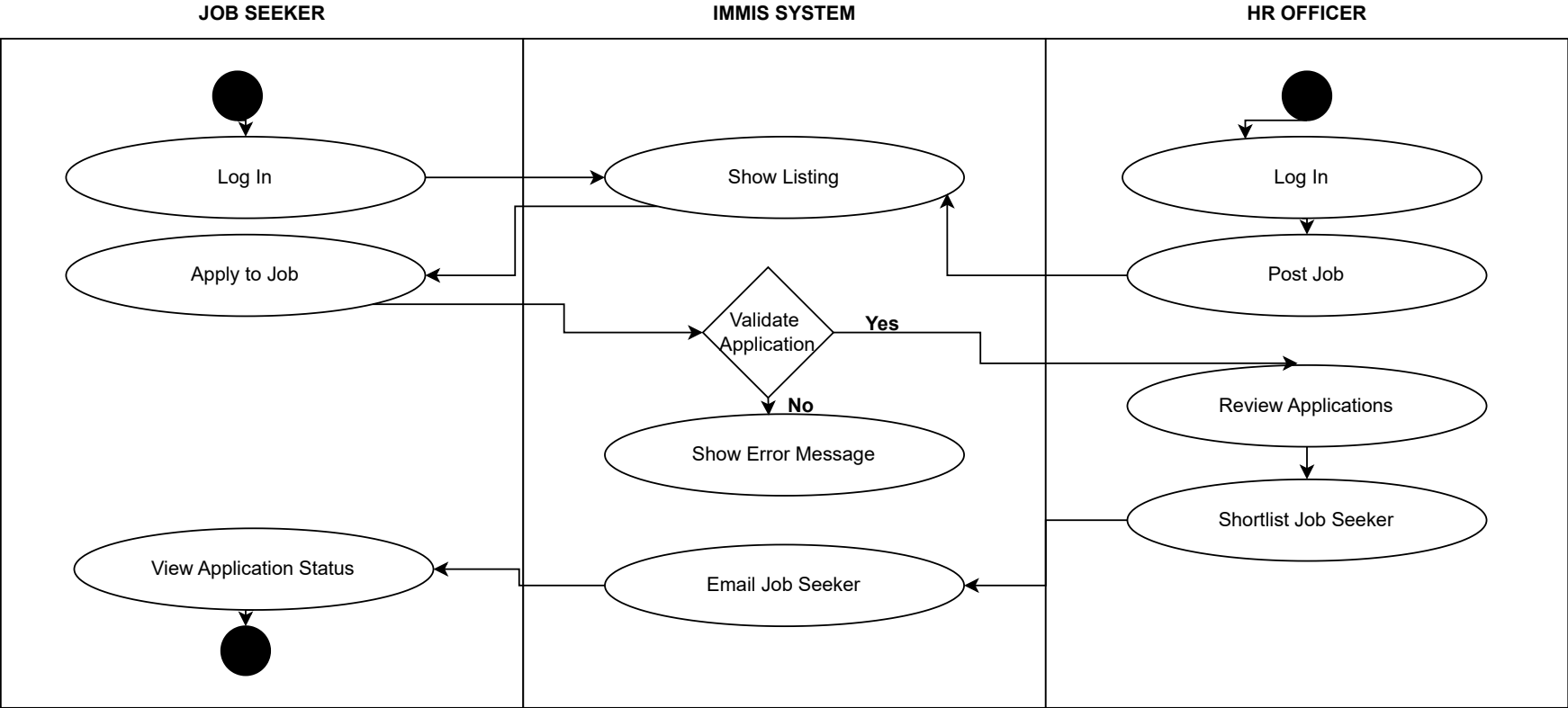
3. TAX Management Service



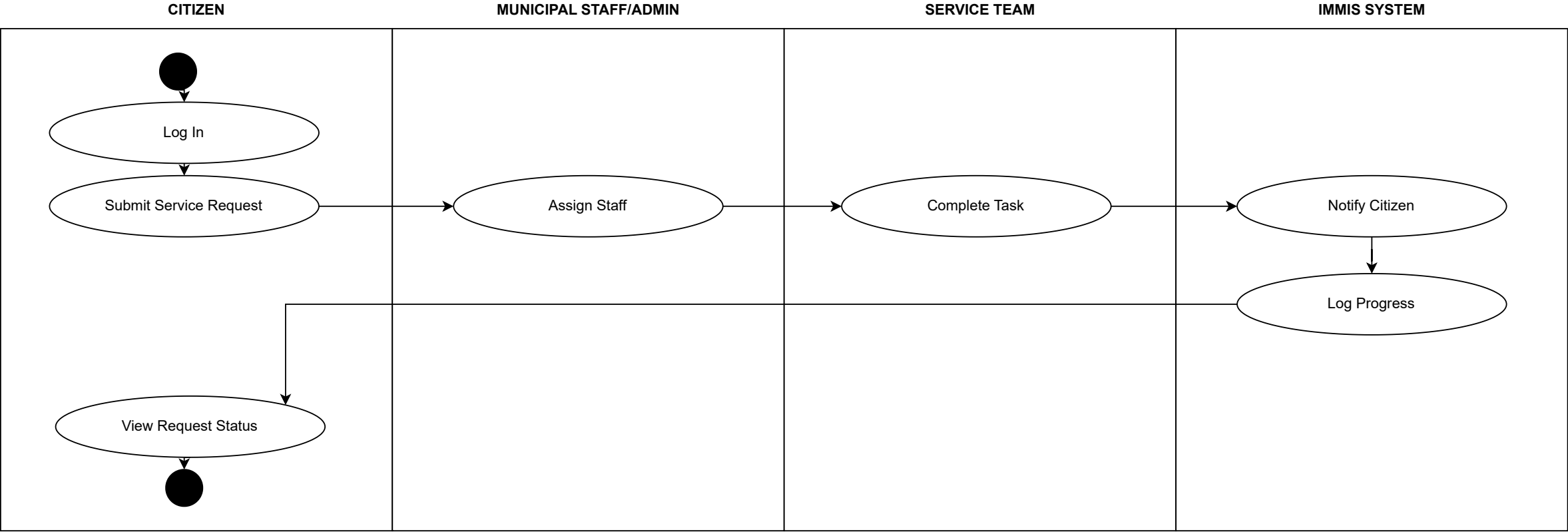
4. TENDER Management Service



5. Job Portal

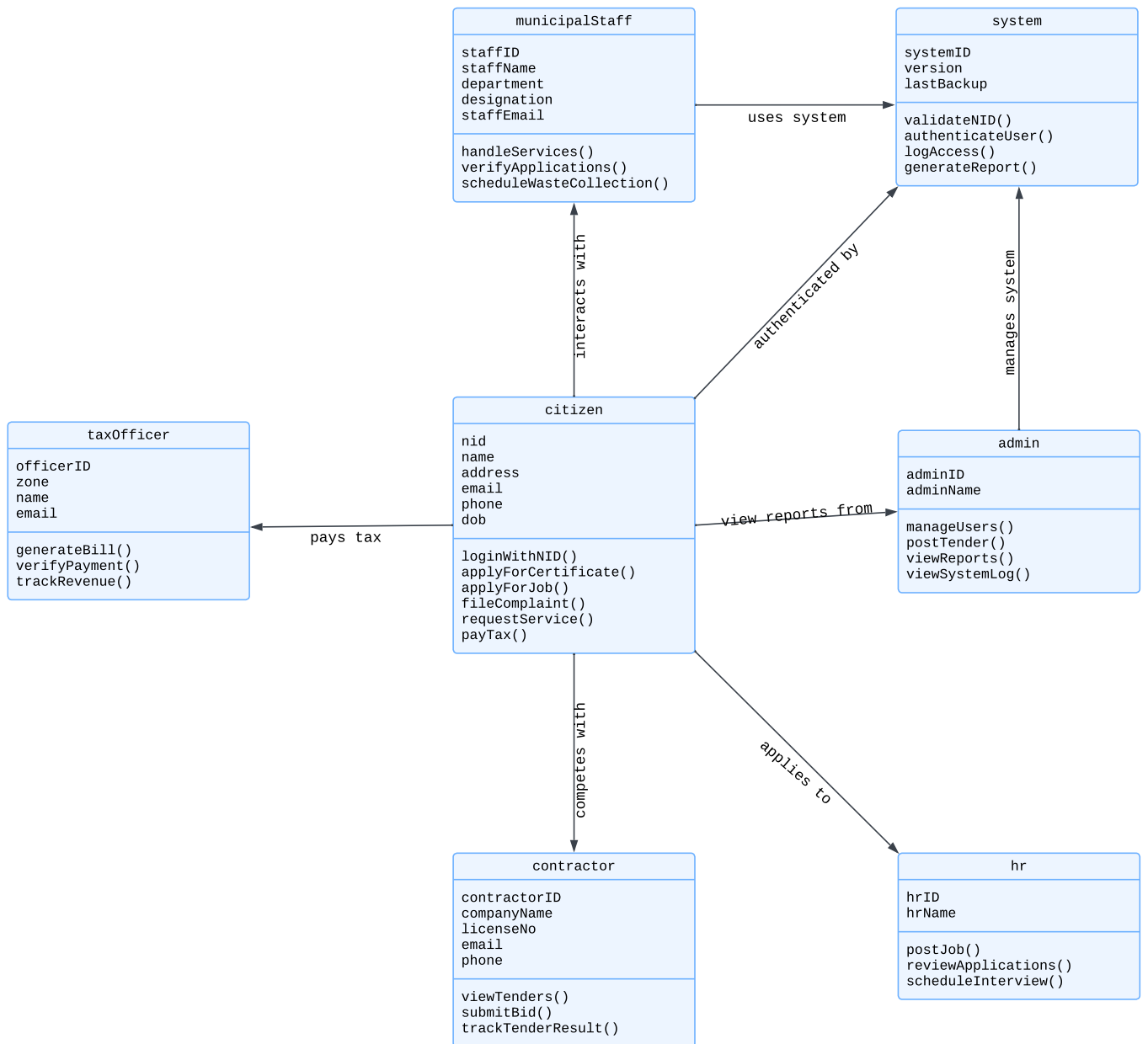


6. Service Request



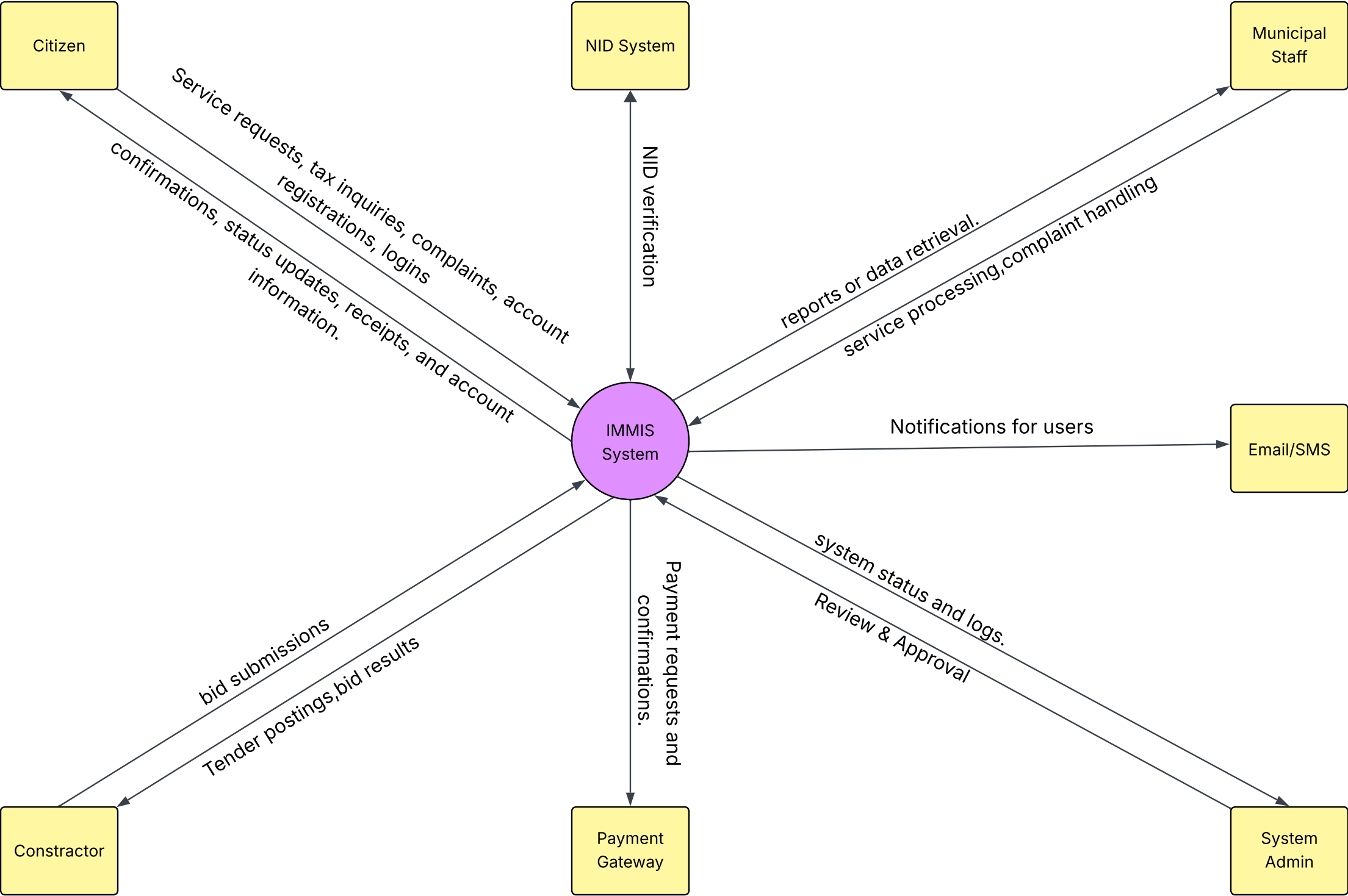
Class Diagram Diagram

CLASS DIAGRAM

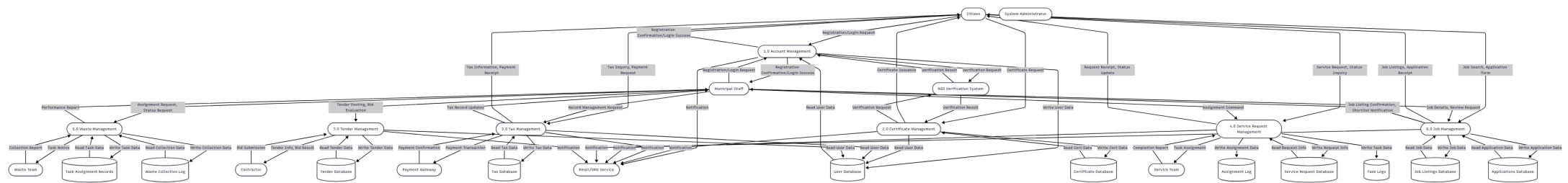


Data Flow Diagram

Level 0: IMMIS System



Level 1: IMMIS System



‘Level 1’ Clear Diagram

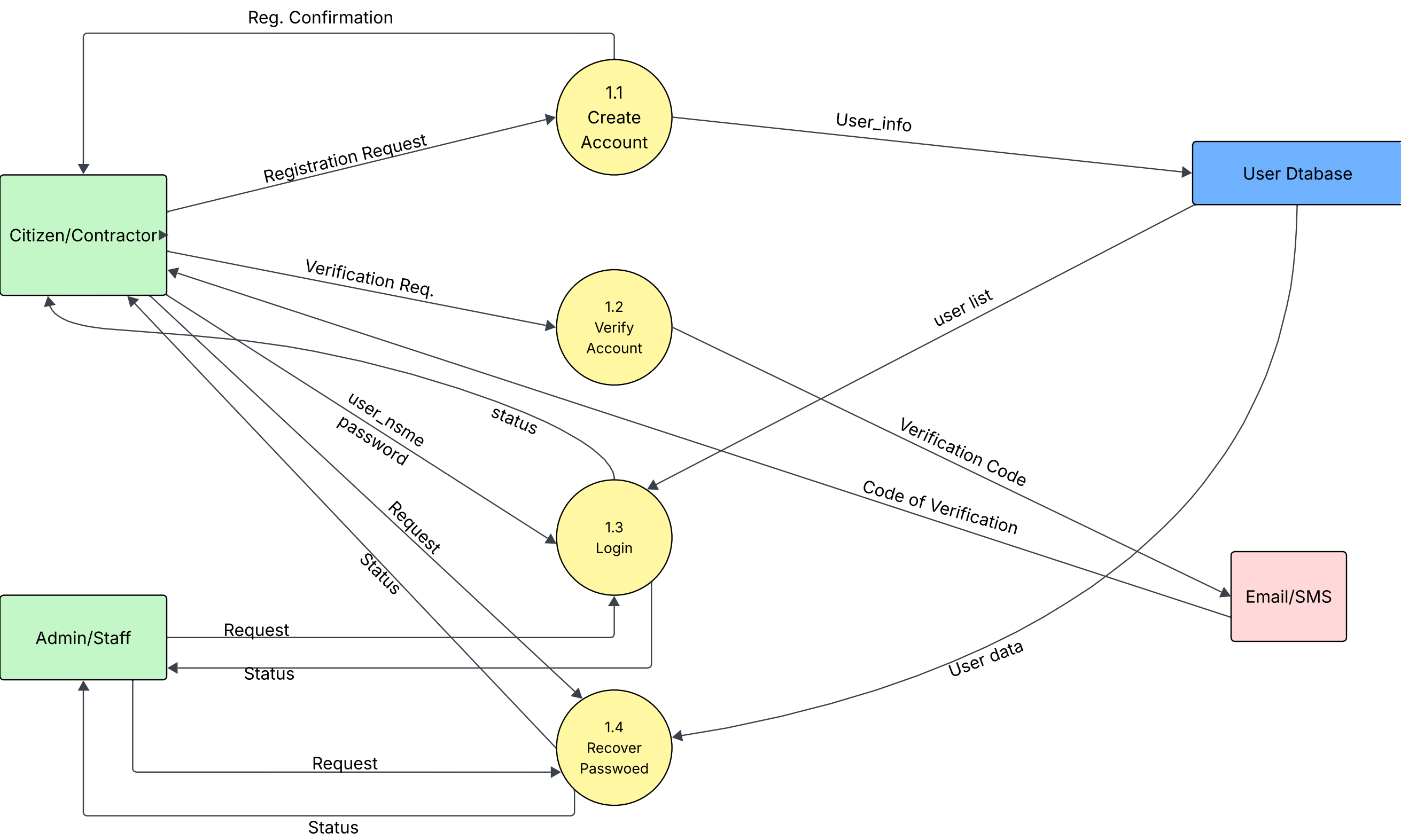
For Clear View of “Level 1” , please download the image(Level-1_DFD_IMMIS.svg) from the given link.

Link:

https://drive.google.com/file/d/1ZrGvw32GNlds5ckhZDwUES5XLMoTj9l_/view?usp=sharing

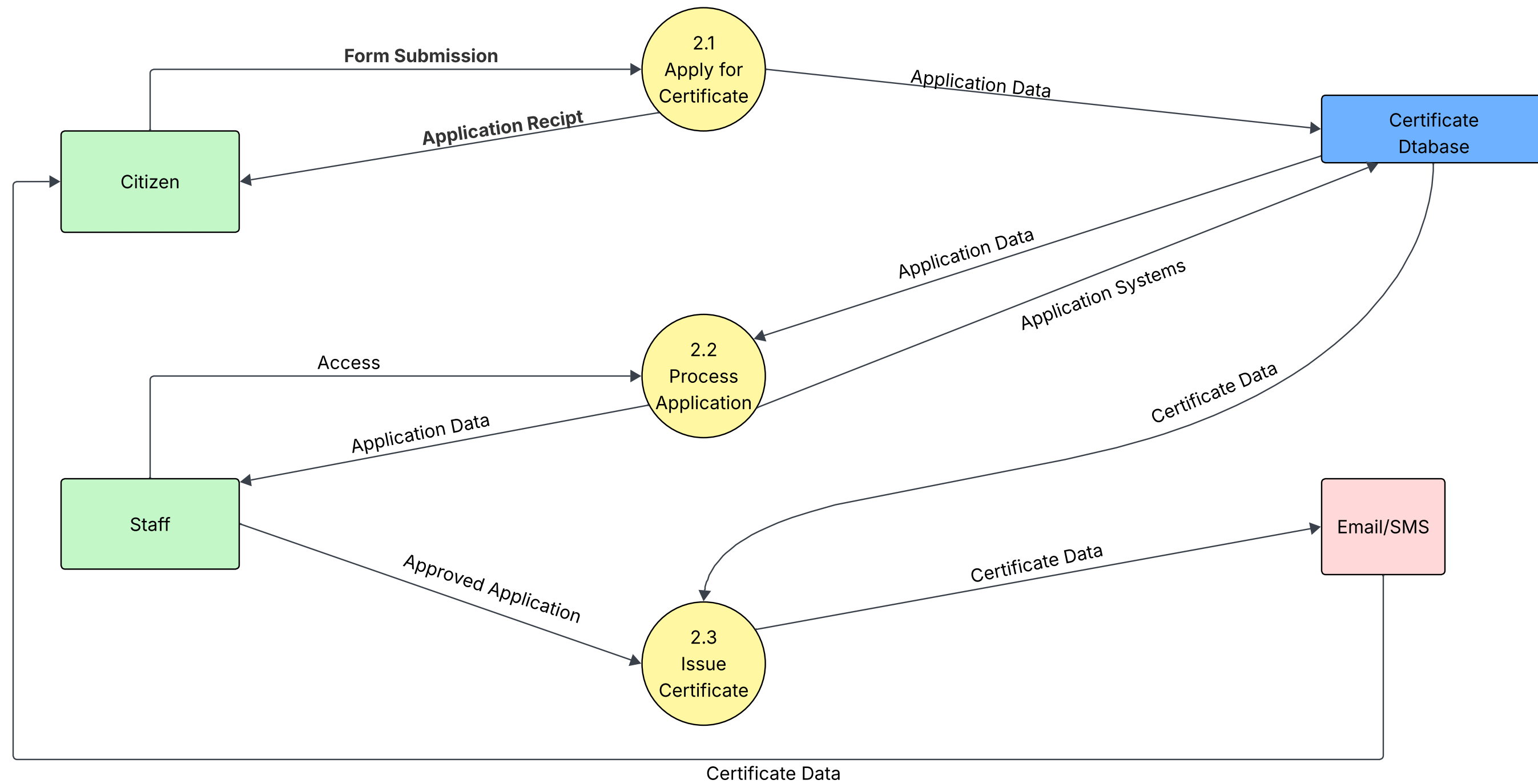
Level 2

1.0 Account Mangement



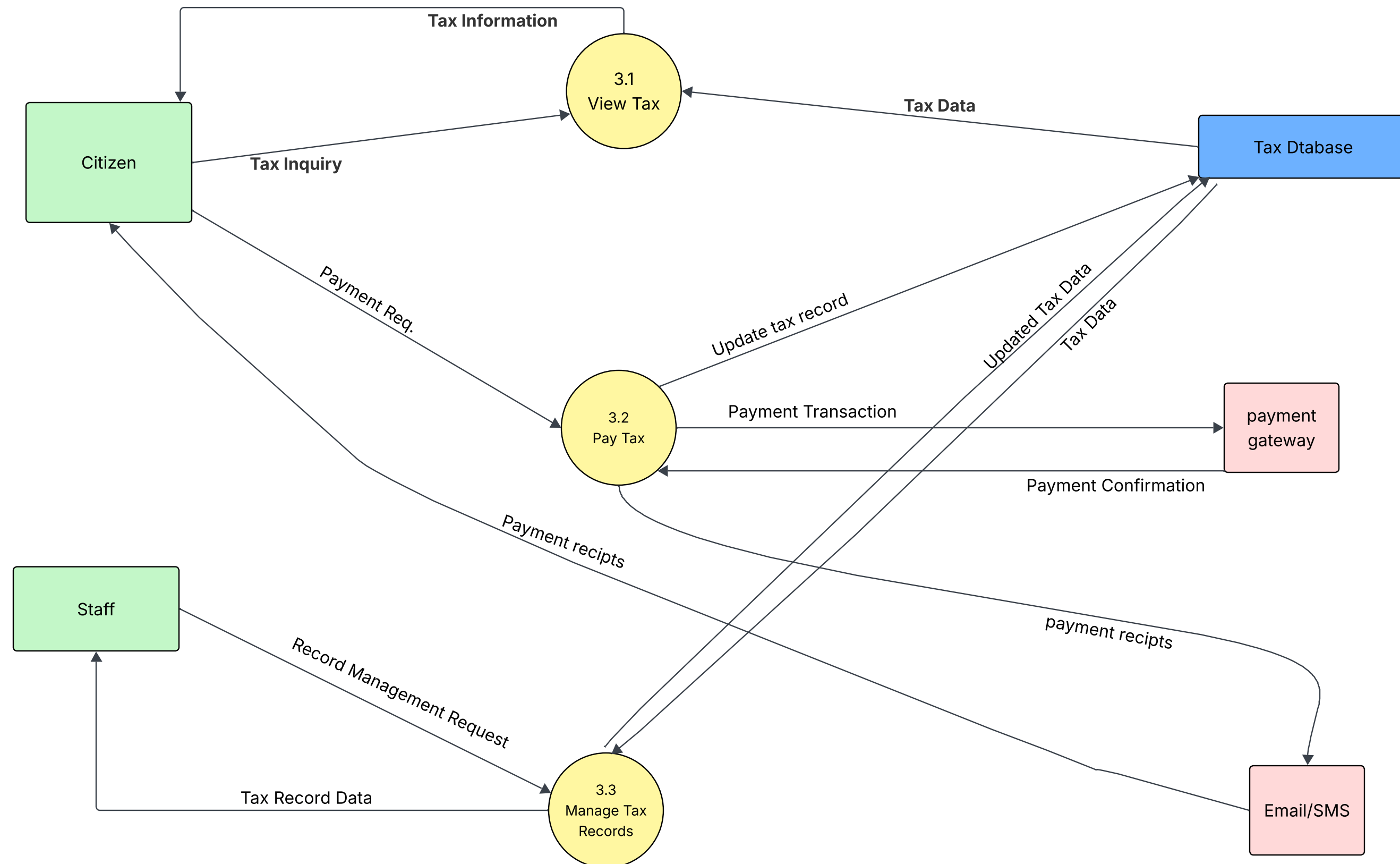
Level 2

2.0 Certificate Mangement



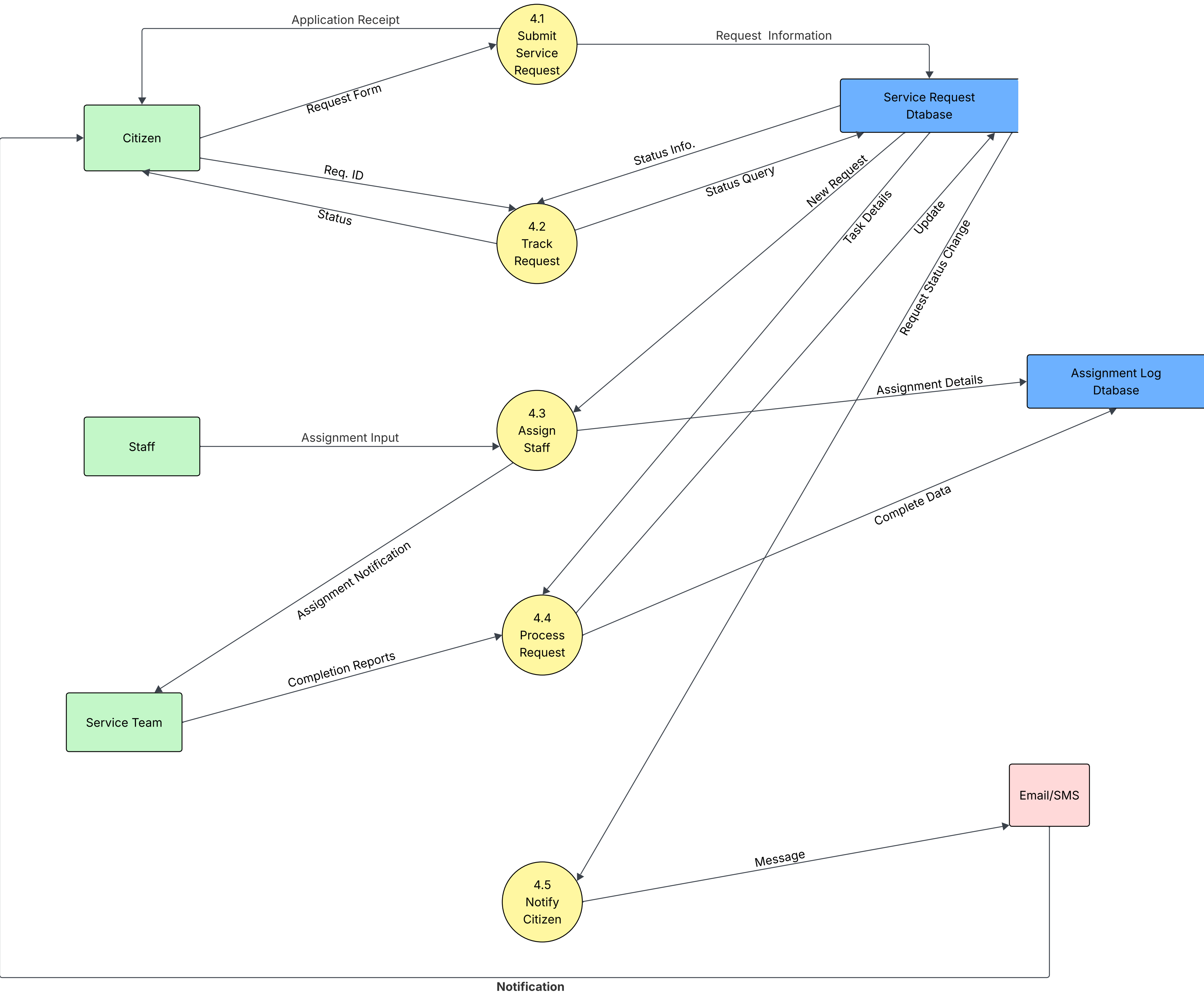
Level 2

3.0 Tax Mangement



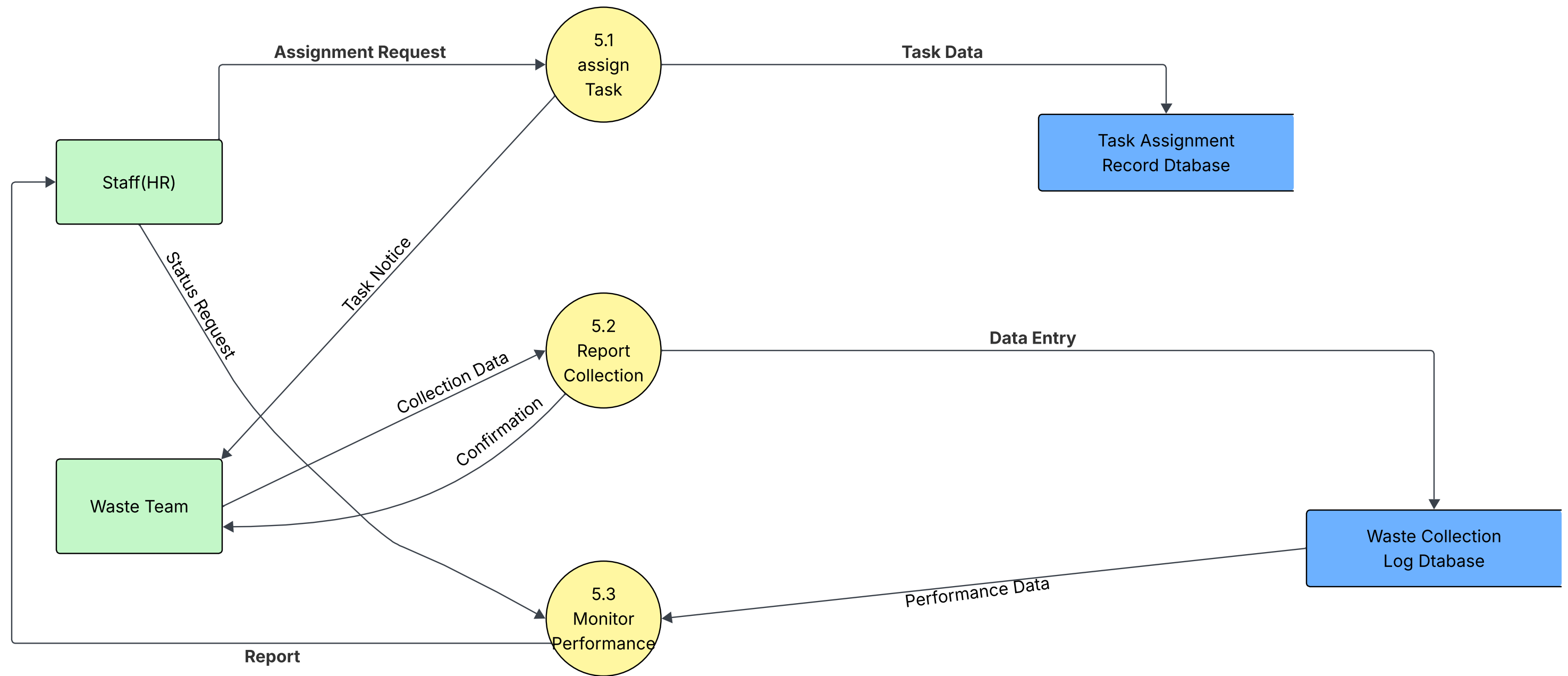
Level 2

4.0 Service Request Mangement



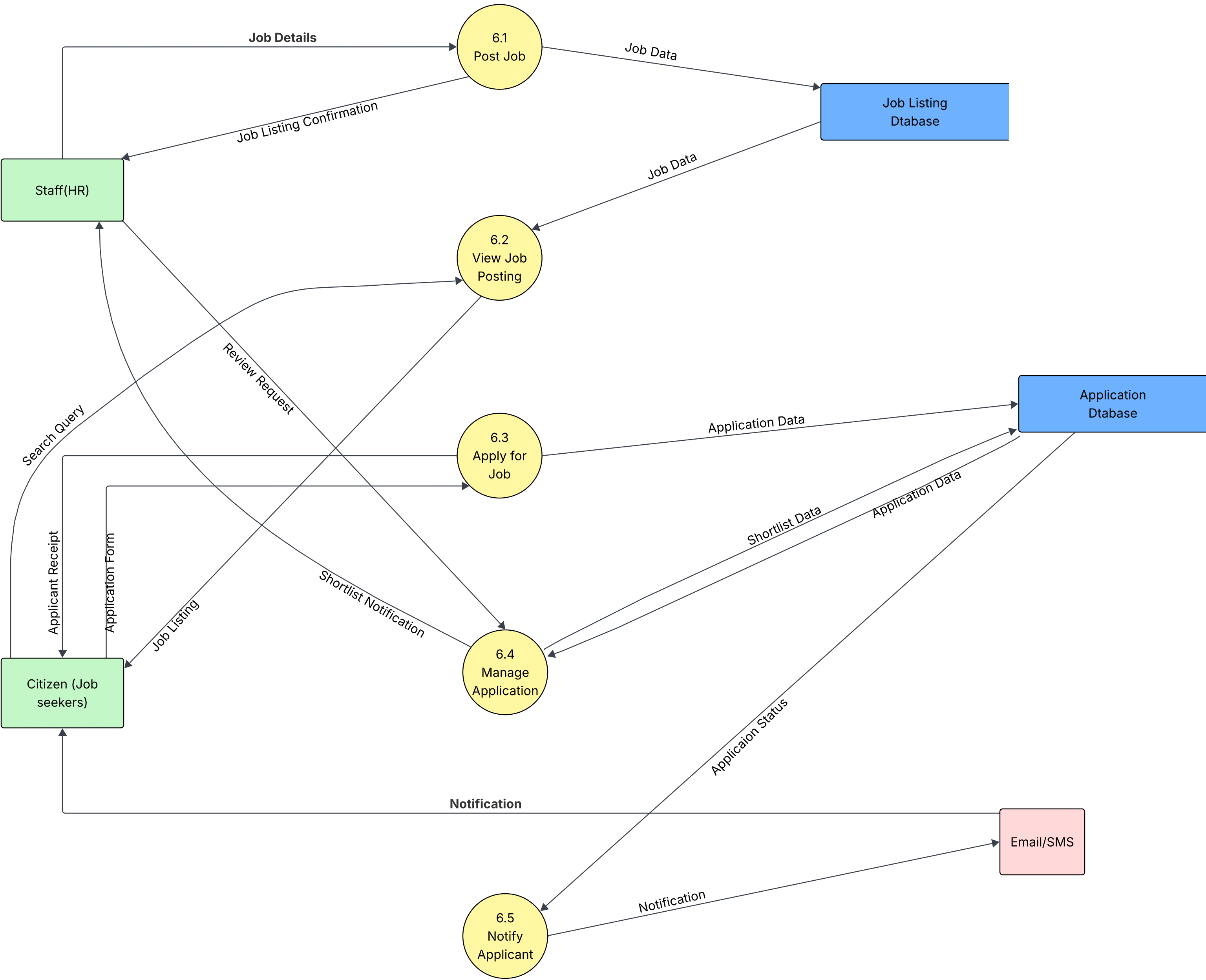
Level 2

5.0 Waste Mangement



Level 2

6.0 Job Mangement



Level 2

7.0 Tender Mangement

